UNIVERSITÀ DEGLI STUDI FIRENZE
Da un secolo, oltre.

HR EXCELLENCE IN RESEARCH

02.3: The Cloud-to-Edge Continuum

Network Applications

LM in Ingegneria Informatica

Daniele Tarchi
Dipartimento Ingegneria dell'Informazione

Foundational Technology Enablers

Objective: To introduce the foundational concepts and technologies that enable the flexible, agile, and intelligent network infrastructure required for 6G and beyond.

Lesson 2.1: Software-Defined Networking (SDN)

- **Duration:** 4 Hours
- **Focus:** Understanding the core principles of SDN, including the decoupling of control and data planes, the role of the centralized controller, and how it enables programmatic network management and agility.

Lesson 2.2: Network Functions Virtualization (NFV)

- **Duration:** 4 Hours
- **Focus:** Exploring how network functions are virtualized and run as software on commodity hardware, leading to cost reduction, increased flexibility, and faster service deployment. We'll also examine the synergy between SDN and NFV.

Lesson 2.3: The Cloud-to-Edge Continuum

- **Duration:** 2 Hours
- **Focus:** Examining the distributed computing paradigm that underpins 6G, including the roles of centralized cloud computing and highly distributed edge computing, and how they seamlessly integrate to form a powerful, intelligent infrastructure fabric.

Cloud Computing & Edge Continuum

A Quick Review: Cloud Computing Models

IaaS (Infrastructure as a Service): The foundational layer, providing basic virtualized resources like virtual machines (VMs), storage, and networking. This model gives you the most control.

- Example: Amazon EC2.
- Note: NFV primarily leverages the IaaS model, using virtualized infrastructure to run network functions as software.

PaaS (Platform as a Service): A layer built on top of IaaS that provides a platform for developing, running, and managing applications without the complexity of building and maintaining the underlying infrastructure.

- Example: Google App Engine.

SaaS (Software as a Service): The top layer, delivering a complete, ready-to-use software application to end-users over the internet.

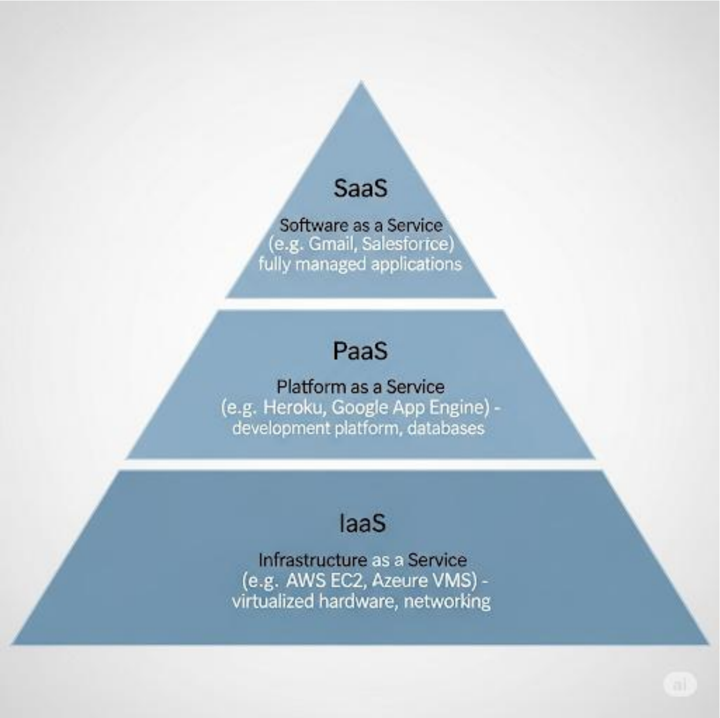
- Example: Gmail, Microsoft 365.

A Quick Review: Cloud Computing Models

IaaS (Infrastructure as a Service): The foundational layer, providing basic virtualized resources like virtual machines (VMs), storage, and networking. While the provider manages the physical infrastructure, the user is responsible for everything from the operating system up, providing maximum control.

Example: Amazon EC2 (Elastic Compute Cloud), Google Compute Engine (GCE), Microsoft Azure Virtual Machines.

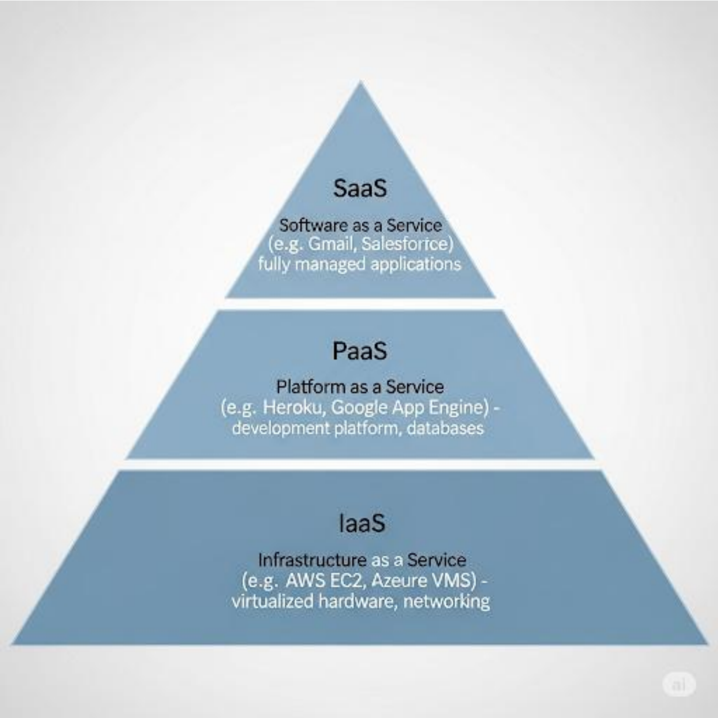
Note: NFV leverages the IaaS model because network operators require granular control over VMs and the networking stack to properly deploy and configure their VNFs.



A Quick Review: Cloud Computing Models

PaaS (Platform as a Service): This layer provides a complete platform for developing and managing applications. The provider handles the underlying infrastructure, operating systems, and runtime environments, allowing the customer to focus solely on their application code.

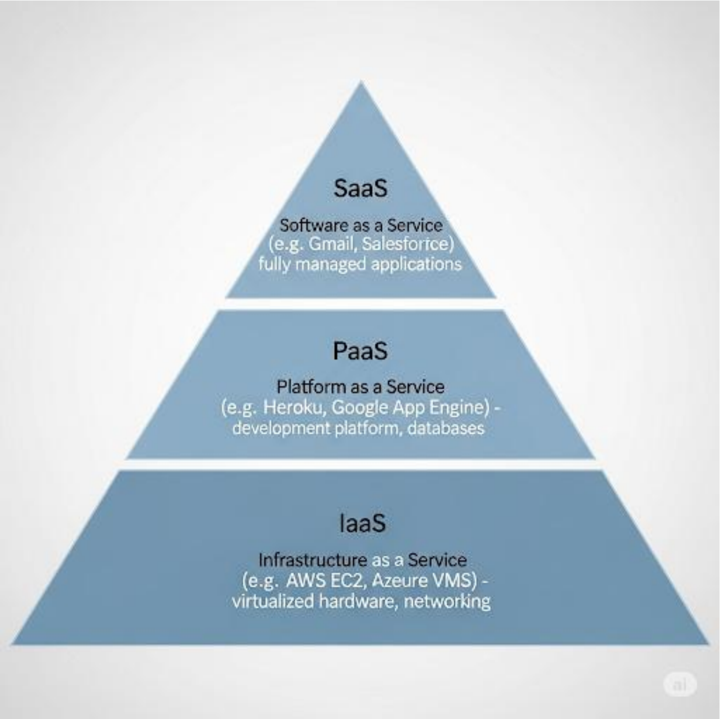
Example: Google App Engine, Heroku, and AWS Elastic Beanstalk.



A Quick Review: Cloud Computing Models

SaaS (Software as a Service): The top layer, delivering a ready-to-use software application directly to end-users over the internet. The provider manages the entire technology stack, and the user only needs a browser to access the service.

Example: Gmail, Microsoft 365, and Salesforce.



The Cloud as a 'Factory' for SDN and NFV

Major cloud computing platforms (like AWS, Azure, and Google Cloud) act as the ideal production environment for large-scale SDN and NFV. The main reason is that a cloud provider's infrastructure, which effectively serves as the **NFVI (Network Functions Virtualization Infrastructure)**, is already a ready-to-use, elastic, and on-demand resource.

In this scenario, key network components like **SDN Controllers** and **NFV Orchestrators** are no longer dedicated hardware; instead, they become software services themselves. They can be run as applications within the cloud, leveraging virtual resources to manage and orchestrate virtualized network functions (VNFs) dynamically and efficiently.

This approach allows companies to:

- **Reduce costs (CapEx):** By replacing proprietary hardware with standard servers and pay-as-you-go cloud services.
- **Increase agility:** By developing and deploying new network services much faster.
- **Scale flexibly:** By adding or removing network resources based on demand, just as you would with any other cloud application.

NFV, SDN, and Cloud: A Symbiotic Relationship

While distinct, **NFV, SDN, and Cloud Computing** are the independent pillars of modern infrastructure. Together, they transition networks from **static hardware silos** to **agile, intelligent platforms** essential for 6G.

NFV: The Virtual Building Blocks

- NFV decouples network functions from proprietary hardware, transforming them into **Virtualized Network Functions (VNFs)**.
- **Hardware Abstraction:** Runs multiple services on standard, shared servers.
- **Rapid Instantiation:** Deploys new services via software in minutes, not weeks.
- **Modular Scaling:** Spin up or down capacity on demand, similar to cloud VMs.

Service Chaining: Modular Intelligence

VNFs act as "Lego blocks" for network architecture. Traffic is directed through a sequential **Service Chain** to apply specific policies and functions automatically.

NFV, SDN, and Cloud: A Symbiotic Relationship

SDN acts as the **intelligent controller**, providing the programmable connectivity needed to link virtual building blocks (VNFs). It replaces manual hardware configuration with high-level software policies.

Dynamic Service Chaining

Instead of static wiring, SDN automatically directs traffic through a specific sequence of services based on user needs:

- **Example**: *Customer Traffic* → **Virtual Firewall** → **DPI** → **Load Balancer**.
- **Benefit:** Changes to the chain are made in software instantly, without touching physical devices.

Enabling 5G/6G Network Slicing

SDN is the key enabler for creating **isolated, virtual networks** on a single physical infrastructure.

- **Customization:** Each "slice" has tailored performance characteristics (latency, bandwidth).
- **Precision:** The SDN data plane ensures traffic is directed precisely to the correct slice and its associated VNFs.

NFV, SDN, and Cloud: A Symbiotic Relationship

Cloud computing provides the essential compute, storage, and networking resources required to run both **VNFs** and the **SDN controller**. It acts as the physical-to-virtual bridge.

From "Peak Provisioning" to "On-Demand"

- **Elastic Scaling:** Resources are provisioned or de-provisioned in real-time based on actual traffic demand.
- **Economic Shift:** Moves network operators from a high-CapEx "provision for peak" model to a flexible **Pay-As-You-Go (OpEx)** model.

The Edge Cloud Revolution (6G Ready)

- Distributed **Edge Clouds** bring processing power closer to the user, which is critical for:
  - **Ultra-Low Latency:** Essential for AR/VR and autonomous vehicles.
  - **Bandwidth Efficiency:** Reduces backhaul traffic by processing data locally.
  - **Global Resilience:** High availability and robust deployment environments.

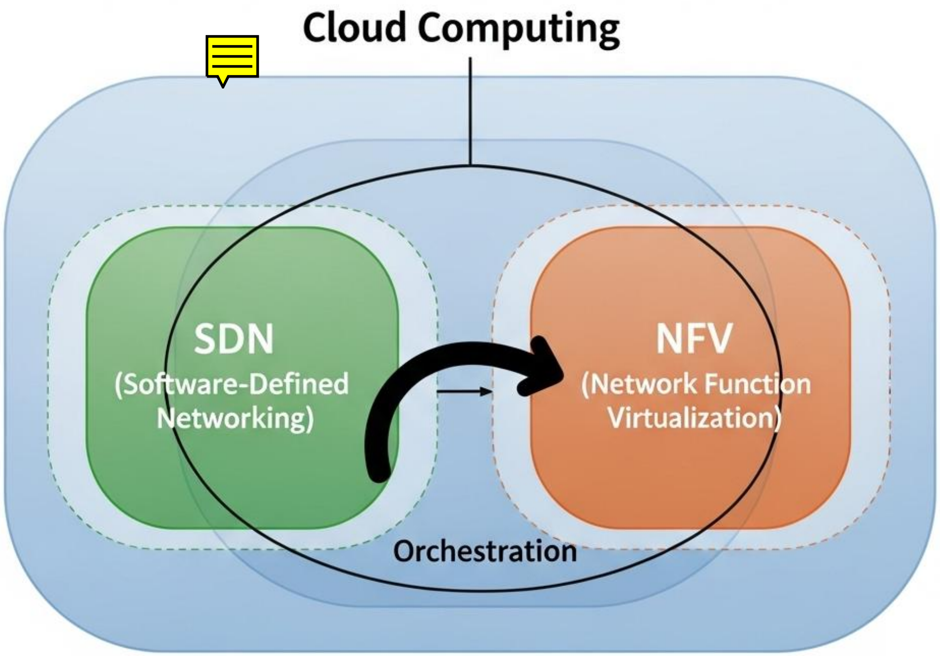
NFV, SDN, and Cloud: A Symbiotic Relationship

In essence

- **Cloud Computing** provides the flexible and scalable underlying hardware infrastructure.
- **NFV** virtualizes the traditional network functions, transforming them into agile software components that can run on this cloud infrastructure.
- **SDN** provides the intelligent, programmable control layer to orchestrate how these virtual functions interact, how traffic flows between them, and how the entire network adapts dynamically to changing demands.

This powerful synergy is what enables the agility, efficiency, and intelligence required for the next generation of mobile networks.

NFV, SDN, and Cloud



The Unbreachable Limit of the Centralized Cloud: Latency

No matter how fast your fiber optics are, the speed of light is finite!

This is the unbreachable limit that real-time applications must confront. However powerful the data centers of a centralized cloud (AWS, Azure, Google), their physical location—often hundreds or thousands of kilometers away from the user—introduces an unavoidable delay.

- **Round-Trip Time (RTT):** The time it takes for a signal to travel the round-trip distance between the user and the server. Even at the speed of light, this time is measurable and significant for applications that require instantaneous responses.

The Cloud's Failure for Real-Time

For critical applications like **self-driving cars**, **augmented reality (AR)**, **telesurgery**, or industrial automation, a delay of even a few tens of milliseconds can be unacceptable or even dangerous. The centralized cloud, because of its inherent latency, cannot meet these requirements. A new paradigm is necessary.

The Solution: Bringing Computation Closer. Edge Computing.

Definition: "A distributed computing paradigm that brings data processing and storage closer to where data is generated and used, to improve response times and save bandwidth."

Edge computing represents a fundamental shift from the centralized cloud model. Instead of relying on distant data centers, it places computational power and storage on devices and local servers situated at the "edge" of the network—right where the action happens.

This new approach effectively overcomes the physical limitations of latency by reducing the distance data needs to travel.

The Solution: Bringing Computation Closer. Edge Computing.

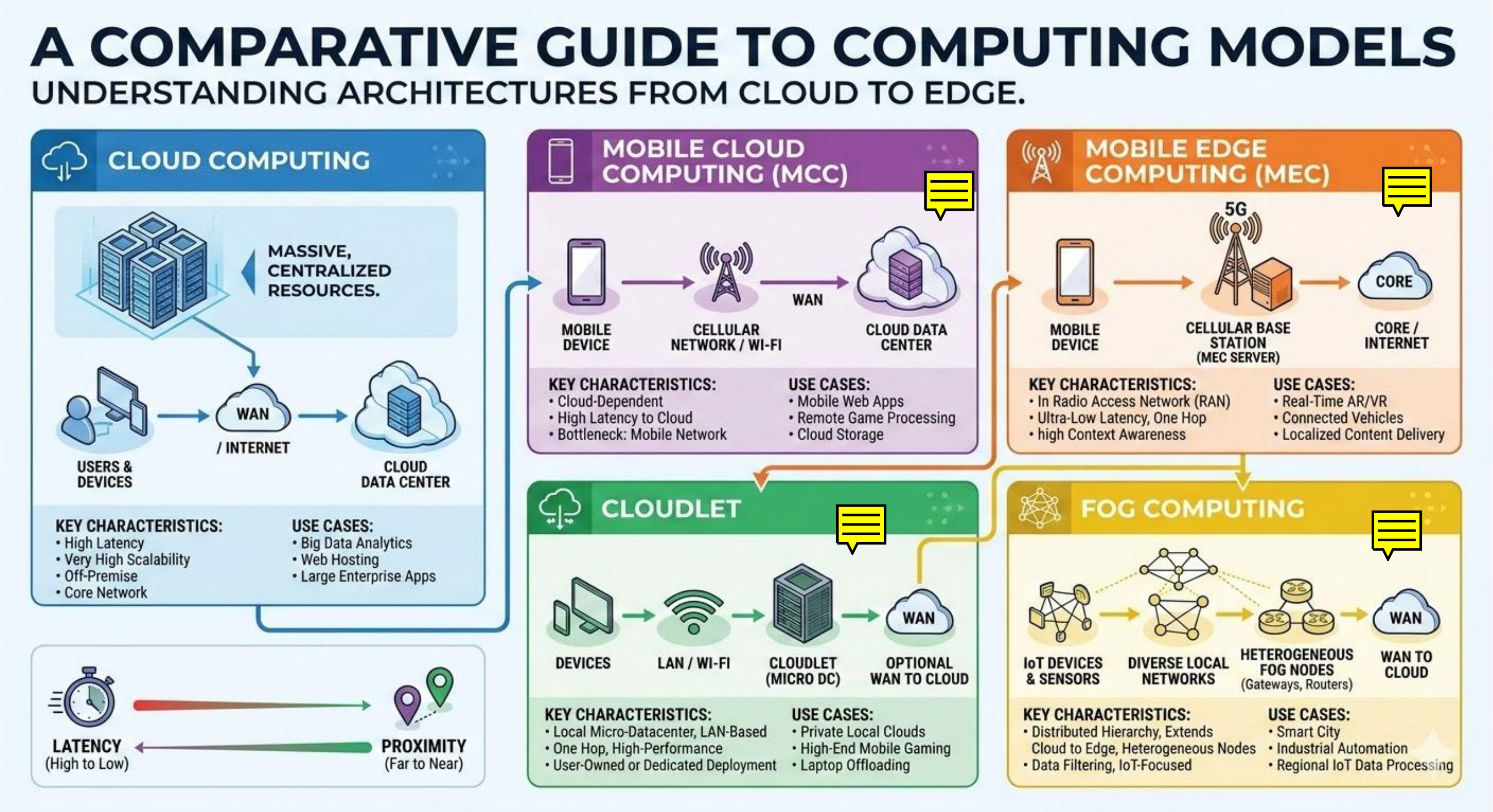
Key Benefits of Edge Computing:

- **Improved Response Times:** By processing data locally, real-time applications receive instant feedback, crucial for autonomous systems and AR/VR experiences.
- **Reduced Bandwidth Usage:** Less data needs to be sent to and from the central cloud, freeing up bandwidth and lowering operational costs.
- **Enhanced Security and Privacy:** Data can be processed and stored locally, reducing the risk of exposure during transmission to the cloud.

This is the paradigm that allows us to build the next generation of applications that were previously impossible.

Da un secolo, oltre.

HR EXCELLENCE IN RESEARCH



Aspect	Cloud Computing	Mobile Cloud Computing	Mobile Edge Computing	Cloudlet	Fog Computing
Definition	Delivery of computing services over the internet.	Combination of cloud and mobile computing to deliver apps and services to mobile devices.	Extends cloud capabilities to the edge of the network, closer to the data source.	Small-scale cloud datacenter located at the edge of the internet.	Decentralized infrastructure placing storage and processing at the network edge.
Primary Use	General-purpose computing, storage, and networking.	Enhancing mobile applications with cloud resources.	Reducing latency and improving performance for mobile applications.	Supporting resource-intensive mobile applications with low latency.	Supporting IoT and real-time applications with low latency and high bandwidth.
Latency	Higher latency due to distance from end-users.	Moderate latency, dependent on cloud infrastructure.	Low latency due to proximity to data source.	Very low latency, typically one wireless hop away.	Low latency due to local processing.
Scalability	Highly scalable with virtually unlimited resources.	Scalable, leveraging cloud infrastructure.	Scalable, but limited by edge node capabilities.	Scalable within the local environment.	Scalable, leveraging both local and cloud resources.

Aspect	Cloud Computing	Mobile Cloud Computing	Mobile Edge Computing	Cloudlet	Fog Computing
Deployment Location	Centralized data centers.	Cloud data centers with mobile access.	Edge of the network, such as base stations or edge nodes.	Edge of the internet, close to mobile devices.	Edge of the network, close to IoT devices and data sources.
Examples	AWS, Google Cloud, Microsoft Azure.	Cloud-based mobile apps, mobile backend as a service (MBaaS).	5G networks, real-time analytics, augmented reality.	Augmented reality, cloud gaming, wearable cognitive assistance.	Smart grids, connected vehicles, industrial IoT.
Advantages	Cost-effective, scalable, flexible, and accessible from anywhere.	Platform-independent, real-time analytics, improved user experience.	High bandwidth, low latency, real-time processing, and decision-making.	Reduced latency, enhanced performance, better battery efficiency.	Reduced latency, bandwidth savings, improved QoS, and edge analytics.
Challenges	Latency, security, data privacy, and dependency on internet connectivity.	Seamless connectivity, security, energy constraints, and vendor lock-in.	Deployment complexity, security, and integration with existing infrastructure.	Implementation complexity, security, and resource management.	Security, data privacy, integration with cloud and edge devices.

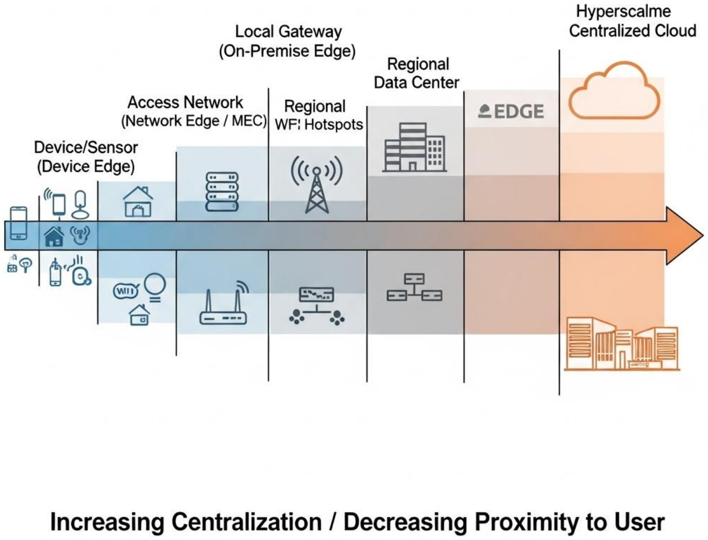
The Cloud-Edge Continuum Spectrum

The **Cloud-Edge Continuum** is a continuous spectrum of computing power and data storage that extends from the user's device all the way to the largest centralized cloud data centers. The key is that these layers are not in a "Cloud vs. Edge" competition, but rather a **collaborative partnership**. Each layer handles tasks best suited for its location and capabilities.

The Layers of the Continuum:

- **Device/Sensor (Device Edge)**
- **Local Gateway (On-Premise Edge)**
- **Access Network (Network Edge / MEC)**
- **Regional Data Center**
- **Hyperscale Centralized Cloud**

The goal is to intelligently distribute computation and data across this spectrum to achieve the optimal balance of speed, cost, and reliability for any given application.



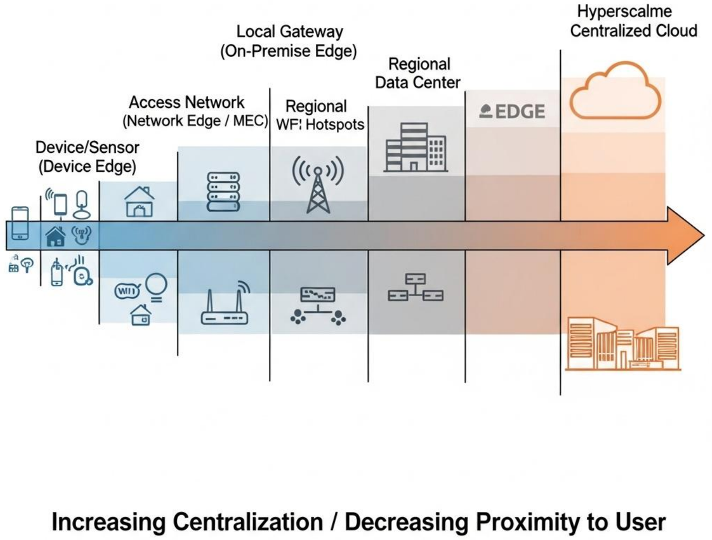
The Cloud-Edge Continuum Spectrum

The Layers of the Continuum:

Device/Sensor (Device Edge): The physical device itself (a smartphone, an AR headset, a car). This is where the initial data is generated and where some local, instant processing can occur.

Local Gateway (On-Premise Edge): A local server or gateway located in the same physical space as the devices, such as a factory floor or a building. It aggregates data and performs low-latency processing for all devices in its immediate vicinity.

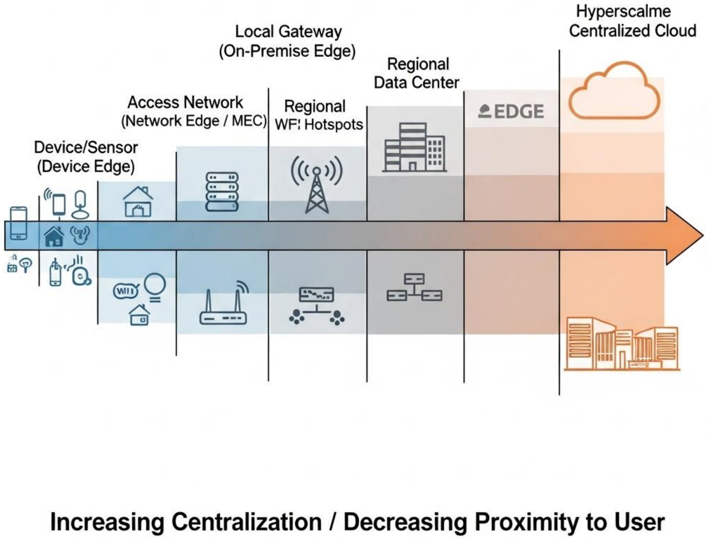
Access Network (Network Edge / MEC): Computing resources located within the network infrastructure, for example, at a 5G cell tower. This layer can serve multiple local gateways and users, providing slightly higher latency but still far less than a centralized cloud.



The Cloud-Edge Continuum Spectrum

Regional Data Center: A smaller data center serving a specific metropolitan or geographic region. It handles aggregated data and services for the surrounding Network Edge locations.

Hyperscale Centralized Cloud: The distant, massive data centers (AWS, Azure, Google Cloud) that provide immense storage, complex analytics, machine learning, and global services. This is where long-term data storage and non-real-time, heavy-duty processing take place.



Why the Edge is Crucial for you

Edge computing is not just a technical trend; it's a critical enabler for the next generation of applications across multiple fields.

Multimedia

Edge is essential for **real-time rendering of AR/VR content**. The headset sends position data to the Edge, which computes and sends back the graphical overlay in just a few milliseconds. This ultra-low latency is the only way to deliver a truly immersive and lag-free experience.

Big Data

Edge allows for **"filtering at the source"**. A factory with thousands of sensors can pre-process data at the Edge and only send aggregated data or anomalies to the cloud. This can **save up to 99% of bandwidth** and reduce storage costs in the central cloud.

Why the Edge is Crucial for you

Advanced Computing

Edge helps solve the complex problem of **hybrid orchestration**. Developers must decide which parts of an application should run on the Edge and which should run on the Cloud to optimize for both performance and cost. Edge provides the flexibility to make these decisions on a case-by-case basis.

Computing Systems and Networks

The Edge is the new frontier for **deploying Virtual Network Functions (VNF) and Cloud-native Network Functions (CNF)**. By placing these functions at the network's edge, service providers can offer innovative, low-latency services directly to users, such as network slicing and enhanced security features.

Comparison Table: When to Use Edge vs. Cloud?

This table clarifies the key differences between Edge and Cloud computing, helping you decide which is the right choice for a given task. They are not mutually exclusive, but rather complementary, and often work together.

Feature	Ideal for the Edge	Ideal for the Cloud
Latency	Very low (5–20 ms)	High (>100 ms)
Bandwidth	Bandwidth savings	Requires high bandwidth
Processing	Real-time, rapid	Intensive, non-real-time, Big Data Analytics
Context	Location-aware	Global and aggregated view
Example	Facial recognition for access	Training the facial recognition model

The Grand Challenge: Edge Orchestration

To introduce the unique management challenges of the edge, relevant for all tracks.

Orchestrating the Edge is not the same as orchestrating the Cloud. While both require sophisticated management, the Edge introduces a new set of critical problems that demand innovative solutions.

Key Challenges

- **Massive Scale**: Instead of managing a few large, centralized data centers, you must now manage tens of thousands of small, geographically distributed edge locations.
- **Resource Constraints**: Edge locations are often limited in power, space, and compute capacity, making resource allocation and optimization much more complex.
- **Intermittent Connectivity**: Edge sites may have less reliable network connections back to the central orchestrator. They must be able to operate autonomously when disconnected.
- **Heterogeneity**: The hardware and software environment can vary significantly across different edge sites, requiring a flexible and adaptive orchestration strategy.

Data Management in the Continuum

To discuss the critical concepts of data consistency and locality in the edge-cloud model.

In the distributed continuum of edge and cloud computing, managing data becomes a key challenge. It's not just about where the data is stored, but how it is processed and kept consistent across the entire system.

Key Concepts

- **Data Locality is Key**: The fundamental principle is to process data as close to its source as possible. This minimizes latency and reduces the need to send large amounts of raw data over the network, saving bandwidth.
- **The Consistency Problem**: A major challenge is managing a shared state that needs to exist at both the edge and the cloud. The question is, how do you ensure that data viewed at the edge is synchronized with the central source in the cloud?

Data Management in the Continuum

An Example: Consider a user's profile. A master copy is stored in the central cloud, but a local cache is maintained at the edge for very fast access.

- **Strong Consistency**: Every read operation returns the most recently written data. This is difficult and expensive to achieve in a distributed system, especially with intermittent connectivity.
- **Eventual Consistency**: Data will eventually become consistent across all locations, but there may be a delay. This is often the more practical approach for the edge, where low latency is prioritized over immediate global consistency.

The Full Software-Defined Infrastructure Stack

To provide a single, comprehensive diagram that ties together every concept from Module 2 into a single, cohesive view.

This diagram represents the complete **Software-Defined Infrastructure Stack**, showing how all the individual components we've discussed work together to create a flexible, programmable, and highly automated network. Think of it as a powerful, multi-layered cake.

The Full Stack

- **Layer 5: The Brain:** The **NFV MANO Framework**
- **Layer 4: The Connectivity:** The **SDN Controller and Data Plane**
- **Layer 3: The Services:** The **VNF and CNF Catalog**
- **Layer 2: The Virtualization Fabric:** The **NFV Infrastructure (NFVI)**
- **Layer 1: The Foundation:** The **Physical Cloud-to-Edge Continuum**

The Full Software-Defined Infrastructure Stack

Layer 5: The Brain The **NFV MANO Framework** orchestrates the entire stack. This is the control plane that manages the lifecycle of network services, from resource allocation and deployment to monitoring and scaling.

Layer 4: The Connectivity The **SDN Controller and Data Plane** provide the programmable, on-demand connectivity. This layer enables the creation of dynamic **Service Function Chains (SFCs)** by linking different network functions together as needed.

Layer 3: The Services The **VNF and CNF Catalog** represents the library of virtual and containerized network functions. These are the building blocks—the firewall, the load balancer, the router—that run on the virtualization fabric.

Layer 2: The Virtualization Fabric The **NFV Infrastructure (NFVI)**, powered by hypervisors and container runtimes (like Docker), creates a unified pool of virtual resources from the underlying hardware.

Layer 1: The Foundation The **Physical Cloud-to-Edge Continuum** is the base hardware layer. It includes everything from massive data centers (the Cloud) to tiny, resource-constrained devices at the edge.

Final Summary & Bridge to Module 3

Module 2 Recap

We have deconstructed the traditional network and rebuilt it from the ground up based on software principles. We now have a virtualized, programmable, and distributed infrastructure fabric that spans from the central cloud to the network edge.

Bridge to Module 3

Now that we have this incredibly powerful and flexible platform, what can we build with it? In Module 3, we will explore the premier applications and paradigms that are enabled by this software-defined infrastructure. We will start by leveraging the 'Network Edge' to enable a new class of applications with **Mobile Edge Computing** and then learn how to provide end-to-end service guarantees over this fabric using **Network Slicing**.

Numerical Examples

5G Interactive Cloud Gaming

The Goal: Determine where to place the game-rendering VNF to ensure a seamless user experience.

The Problem: In cloud gaming, graphics are rendered in the network and streamed to the device. Any delay between a button press and the visual update (input lag) ruins the gameplay.

The Target Service: A competitive "First Person Shooter" (FPS).

The Hard Constraint: To maintain "tactile" responsiveness, the **Total Network Latency** (L_{total}) must be:

$$L_{total} < 20ms$$

The Formula for End-to-End Latency

To calculate the actual delay, we model the round-trip path using this formula:

$$L_{total} = (2 \times L_{prop}) + (2 \times L_{net}) + L_{proc}$$

Variable Definitions:

- L_{prop} (**Propagation Delay**): The time for the signal to travel the physical distance through fiber.
  - Standard Constant: Approx. $5\mu s$ **per km**.
- L_{net} (**Network Delay**): The cumulative time spent in routers/switches for queuing and switching.
  - Assumption: $1ms$ **per hop**.
- L_{proc} (**Processing Delay**): The time the VNF takes to render the frame or compute the game logic.

Comparing Two Infrastructure Tiers

We must choose between hosting the gaming VNF in a regional data center or at the local cell site.

Parameter	Tier A: Central Cloud	Tier B: Edge Computing
Distance (One-way)	500km	20km
Network Hops	6 Hops	2 Hops
Processing (L_{proc})	2ms (High-perf Server)	8ms (Resource-constrained)
Cost	Low (Economy of scale)	High (Distributed Hardware)

Step 1 — Calculating the Central Cloud (Tier A)

Let's see if the cheaper, more powerful central cloud can handle the service.

1. **Calculate Propagation Delay (L_{prop})**:

$$500km \times 0.005ms/km = 2.5ms$$

2. **Apply the Total Latency Formula**:

$$L_{total} = (2 \times 2.5ms) + (2 \times 6ms) + 2ms = 5ms + 12ms + 2ms$$

   - **Result**: $L_{total} = 19ms$

Conclusion: It meets the requirement ($19ms < 20ms$), but with **zero safety margin**. Any minor network congestion will cause the service to fail.

Step 2 — Calculating the Edge Computing (Tier B)

Now we look at the edge, where compute power is lower but physical distance is shorter.

1. **Calculate Propagation Delay (L_{prop})**:
$$20km \times 0.005ms/km = 0.1ms$$
2. **Apply the Total Latency Formula**:
$$L_{total} = (2 \times 0.1ms) + (2 \times 2ms) + 8ms = 0.2ms + 4ms + 8ms$$
   - **Result**: $L_{total} = 12.2ms$

Conclusion: It safely meets the requirement ($12.2ms < 20ms$). We have a $7.8ms$ "**buffer**" to handle network jitter or rendering spikes.

Step 3 — Discussion: The Cost-Performance Trade-off

Summary of Findings:

- EC is the clear winner for performance: 12.2ms vs 19ms.
- Cloud is the winner for cost: It uses cheaper, high-density hardware in a single location.

Question: If the server load increases and the L_{proc} fluctuates by only +2ms, what happens to the user experience in Tier A vs Tier B?

- Answer:
  - Tier A (Cloud): 19ms + 2ms = 21ms. FAIL. (Latency exceeds the 20ms threshold).
  - Tier B (EC): 12.2ms + 2ms = 14.2ms. PASS. (Still well within the acceptable limit).

Final Takeaway: EC isn't just about being "faster"; it is about providing the reliability required for ultra-low latency SLAs.

UNIVERSITÀ DEGLI STUDI FIRENZE
Da un secolo, oltre.

HR EXCELLENCE IN RESEARCH

02.3: The Cloud-to-Edge Continuum

Network Applications

LM in Ingegneria Informatica

Daniele Tarchi
Dipartimento Ingegneria dell'Informazione
daniele.tarchi@unifi.it